

Lecture Slide – 8 LU Decomposition

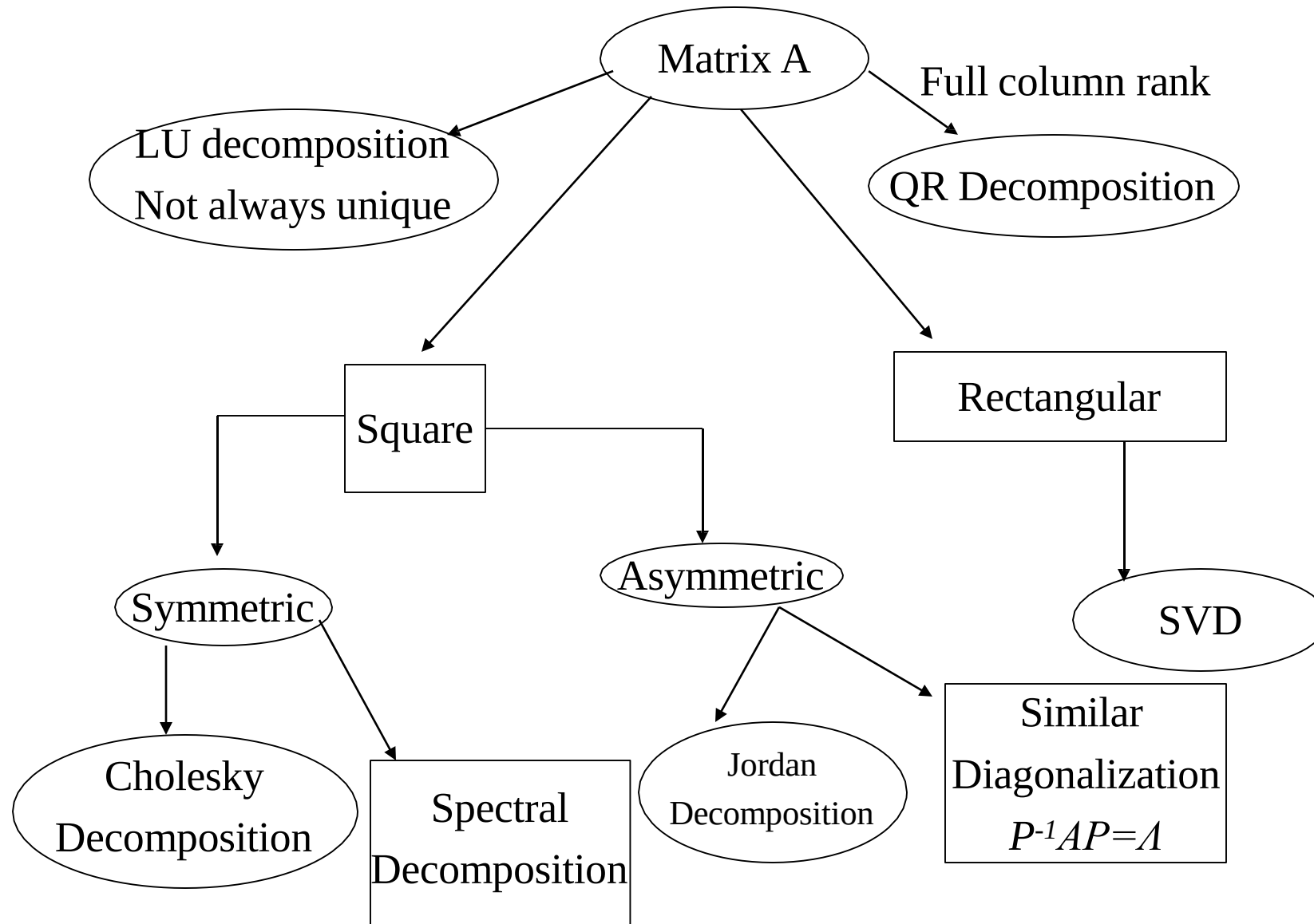
Matrix Decomposition

Gaussian elimination is a direct method for solving a system of linear equations $Ax=b$. It transforms the matrix A into an upper triangular form and then uses back substitution.

Problem: If you need to solve **multiple systems with the same coefficient matrix A** but with different right-hand side vectors b , Gaussian elimination becomes inefficient because you must repeat the elimination steps every time.

- Decompositions provide a numerically stable way to solve a system of linear equations, and
- To invert a matrix.
- Additionally, they provide an important tool for analyzing the numerical stability of a system.

Decomposition in Diagram



Easy to solve system

Upper Triangular Matrix: Used in Gauss Elimination Method

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ 0 & a_{22} & a_{23} & \dots & a_{2n} \\ 0 & 0 & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{bmatrix}$$

Solution: This system is solved using Backward substitution

$$x_i = (b_i - \sum_{j=i+1}^n a_{ij} \cdot x_j) / a_{ii}$$

Easy to solve system

Lower triangular matrix: Can be used also

$$\begin{bmatrix} a_{11} & 0 & 0 & \dots & 0 \\ a_{21} & a_{22} & 0 & \dots & 0 \\ a_{31} & a_{32} & a_{33} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{bmatrix}$$

Solution: This system is solved using forward substitution

$$x_i = (b_i - \sum_{j=1}^{i-1} a_{ij} \cdot x_j) / a_{ii}$$

Easy to solve system

Some linear system that can be easily solved

$$\begin{bmatrix} a_{11} & 0 & 0 & \dots & 0 \\ 0 & a_{22} & 0 & \dots & 0 \\ 0 & 0 & a_{33} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{bmatrix}$$

The solution:

$$\begin{bmatrix} b_1 / a_{11} \\ b_2 / a_{22} \\ \vdots \\ b_n / a_{nn} \end{bmatrix}$$

LU Decomposition

Let A be a $m \times m$ with nonsingular square matrix. Then there exists two matrices L and U such that, where L is a lower triangular matrix and U is an upper triangular matrix.

$$U = \begin{bmatrix} u_{11} & u_{12} & \cdots & u_{1m} \\ 0 & u_{22} & \cdots & u_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{mm} \end{bmatrix} \quad \text{and} \quad L = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ l_{m1} & l_{m2} & \cdots & l_{mm} \end{bmatrix}$$

Where, $A = LU$

- Once $A=LU$ is computed, solving $Ax=b$ reduces to:
- First solve $Ly=b$ using **forward substitution**,
- Then solve $Ux=y$ using **back substitution**.
- The decomposition $A=LU$ needs to be computed **only once**.
- For different right-hand side vectors b , you can reuse L and U without repeating elimination steps.

Calculation of L and U (cont.)

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 12 & -8 & 6 \\ 3 & -13 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ 12 & -8 & 6 \\ 3 & -13 & 2 \end{bmatrix}$$

Now reducing the first column we have

$$\begin{bmatrix} 6 & -2 & 2 \\ 0 & -4 & 2 \\ 0 & -12 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1/2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ 12 & -8 & 6 \\ 3 & -13 & 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 6 & -2 & 2 \\ 0 & -4 & 2 \\ 0 & 0 & -5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1/2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ 12 & -8 & 6 \\ 3 & -13 & 2 \end{bmatrix}$$

Calculation of L and U (cont.)

$$\text{Now } \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1/2 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1/2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1/2 & 3 & 1 \end{bmatrix}$$

Therefore,

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 12 & -8 & 6 \\ 3 & -13 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1/2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ 0 & -4 & 2 \\ 0 & 0 & -5 \end{bmatrix} = LU$$

If A is a Non singular matrix then for each L (lower triangular matrix) the upper triangular matrix is unique but an LU decomposition is not unique. There can be more than one such LU decomposition for a matrix. Such as

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 12 & -8 & 6 \\ 3 & -13 & 2 \end{bmatrix} = \begin{bmatrix} 6 & 0 & 0 \\ 12 & 1 & 0 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{2}{6} & \frac{2}{6} \\ 0 & -4 & 2 \\ 0 & 0 & -5 \end{bmatrix} = LU$$

Calculation of L and U (cont.)

- Thus LU decomposition is not unique. Since we compute LU decomposition by elementary transformation so if we change L then U will be changed, such that $A=LU$
- To find out the unique LU decomposition, it is necessary to put some restriction on L and U matrices. For example, we can require the lower triangular matrix L to be a unit one (i.e. set all the entries of its main diagonal to ones).
- Uses: The LU decomposition is most commonly used in the solution of systems of simultaneous linear equations. We can also find determinant easily by using LU decomposition (Product of the diagonal element of upper and lower triangular matrix).

Solving system of linear equation using LU decomposition

- Suppose we would like to solve a $n \times n$ system $AX = b$. Then we can find a LU-decomposition for A, then to solve $AX = b$, it is enough to solve the systems

$$LY = b$$

$$UX = Y$$

- Thus the system $LY = b$ can be solved by the method of forward substitution and the system $UX = Y$ can be solved by the method of backward substitution. To illustrate, we give some examples
- Consider the given system $AX = b$, where

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 12 & -8 & 6 \\ 3 & -13 & 2 \end{bmatrix} \quad \text{and} \quad b = \begin{pmatrix} 8 \\ 14 \\ -17 \end{pmatrix}$$

Solving system of linear equation using LU decomposition

We have seen $A = LU$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1/2 & 3 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 6 & -2 & 2 \\ 0 & -4 & 2 \\ 0 & 0 & -5 \end{bmatrix}$$

Thus, to solve $AX = \mathbf{b}$, we first solve $LY = \mathbf{b}$ by forward substitution

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1/2 & 3 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 14 \\ -17 \end{bmatrix} \quad \text{Then} \quad Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 8 \\ -2 \\ -15 \end{bmatrix}$$

Solving system of linear equation using LU decomposition

Now, we solve $UX = Y$ by backward substitution

$$\begin{bmatrix} 6 & -2 & 2 \\ 0 & -4 & 2 \\ 0 & 0 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ -2 \\ -15 \end{bmatrix} \quad \text{then} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Method: [A] Decomposes to [L] and [U]

$$[A] = [L][U] = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

- [U] is the same as the coefficient matrix at the end of the forward elimination step.
- [L] is obtained using the *multipliers* that were used in the forward elimination process

Finding the $[U]$ matrix

Using the Forward Elimination Procedure of Gauss Elimination

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

Step 1: $\frac{64}{25} = 2.56$; $Row2 - Row1(2.56) =$

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 144 & 12 & 1 \end{bmatrix}$$

$$\frac{144}{25} = 5.76; \quad Row3 - Row1(5.76) = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & -16.8 & -4.76 \end{bmatrix}$$

Finding the $[U]$ matrix

Matrix after Step 1:

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & -16.8 & -4.76 \end{bmatrix}$$

Step 2: $\frac{-16.8}{-4.8} = 3.5$; $Row3 - Row2(3.5) =$

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}$$

$$[U] = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}$$

Finding the $[L]$ matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}$$

Using the multipliers used during the Forward Elimination Procedure

From the first step of
forward elimination

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

$$\ell_{21} = \frac{a_{21}}{a_{11}} = \frac{64}{25} = 2.56$$

$$\ell_{31} = \frac{a_{31}}{a_{11}} = \frac{144}{25} = 5.76$$

Finding the $[L]$ matrix

From the second step
of forward elimination

$$\left[\begin{array}{ccc|c} 25 & 5 & 1 & \\ 0 & -4.8 & -1.56 & \\ \hline 0 & -16.8 & -4.76 & \end{array} \right]$$

$$\ell_{32} = \frac{a_{32}}{a_{22}} = \frac{-16.8}{-4.8} = 3.5$$

$$[L] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & \\ 2.56 & 1 & 0 & \\ \hline 5.76 & 3.5 & 1 & \end{array} \right]$$

Does $[L][U] = [A]$?

$$[L][U] = \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} = ?$$

Using LU Decomposition to solve SLEs

Solve the following set of linear equations using LU Decomposition

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Using the procedure for finding the $[L]$ and $[U]$ matrices

$$[A] = [L][U] = \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}$$

Example

$$\text{Set } [L][Y] = [b] \quad \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Solve for $[Y]$

$$y_1 = 106.8$$

$$2.56y_1 + y_2 = 177.2$$

$$5.76y_1 + 3.5y_2 + y_3 = 279.2$$

Example

Complete the forward substitution to solve for $[Y]$

$$y_1 = 106.8$$

$$\begin{aligned} y_2 &= 177.2 - 2.56y_1 \\ &= 177.2 - 2.56(106.8) \\ &= -96.2 \end{aligned}$$

$$\begin{aligned} y_3 &= 279.2 - 5.76y_1 - 3.5y_2 \\ &= 279.2 - 5.76(106.8) - 3.5(-96.21) \\ &= 0.735 \end{aligned}$$

$$[Y] = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ -96.21 \\ 0.735 \end{bmatrix}$$

Example

$$\text{Set } [U][X] = [Y] \quad \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ -96.21 \\ 0.735 \end{bmatrix}$$

Solve for $[X]$

The 3 equations become

$$\begin{aligned} 25x_1 + 5x_2 + x_3 &= 106.8 \\ -4.8x_2 - 1.56x_3 &= -96.21 \\ 0.7x_3 &= 0.735 \end{aligned}$$

Example

From the 3rd equation

$$0.7x_3 = 0.735$$

$$x_3 = \frac{0.735}{0.7}$$

$$x_3 = 1.050$$

Substituting in a_3 and using the second equation

$$-4.8x_2 - 1.56x_3 = -96.21$$

$$x_2 = \frac{-96.21 + 1.56a_3}{-4.8}$$

$$x_2 = \frac{-96.21 + 1.56(1.050)}{-4.8}$$

$$x_2 = 19.70$$

Example

Substituting in x_3 and x_2 using the first equation

$$25x_1 + 5x_2 + x_3 = 106.8$$

$$\begin{aligned} x_1 &= \frac{106.8 - 5x_2 - x_3}{25} \\ &= \frac{106.8 - 5(19.70) - 1.050}{25} \\ &= 0.2900 \end{aligned}$$

Hence the Solution Vector is:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0.2900 \\ 19.70 \\ 1.050 \end{bmatrix}$$

Practice Problem

Derive an LU decomposition for following matrix

$$[A] = \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0.1 & 7 & -0.3 \\ 0.3 & -0.2 & 10 \end{bmatrix}$$

After forward elimination, the following upper triangular matrix was obtained:

$$[U] = \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.0120 \end{bmatrix} \quad f_{21} = \frac{0.1}{3} = 0.0333333 \quad f_{31} = \frac{0.3}{3} = 0.100000$$
$$f_{32} = \frac{-0.19}{7.00333} = -0.0271300$$

Practice Problem

Thus, the lower triangular matrix is

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix}$$

Consequently, the LU decomposition is

$$[A] = [L][U] = \begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix} \begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.0120 \end{bmatrix}$$

Matrix Inverse from LU Decomposition

The inverse can be computed in a **column-by-column fashion** by generating solutions **with unit vectors as the right-hand-side constants**. For example, if the right-hand-side constant has a 1 in the first position and zeros elsewhere,

$$\{b\} = \begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix}$$

The resulting solution will be the first column of the matrix inverse. Similarly, if a unit vector with a 1 at the second row is used

$$\{b\} = \begin{Bmatrix} 0 \\ 1 \\ 0 \end{Bmatrix}$$

The result will be the second column of the matrix inverse.

Matrix Inverse from LU Decomposition

From Previous Example

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix}$$

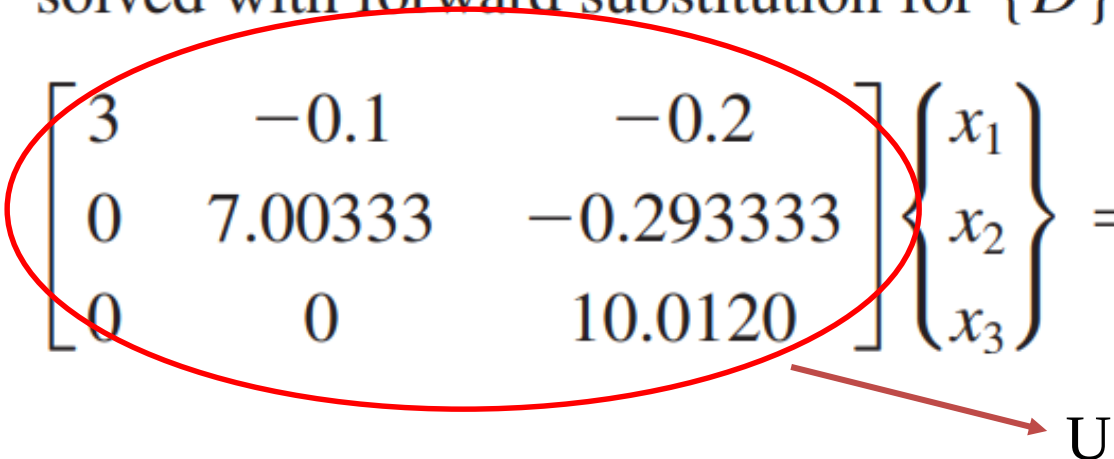
The first column of the matrix inverse can be determined by

$$\begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix} \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} = \begin{Bmatrix} 1 \\ 0 \\ 0 \end{Bmatrix}$$

solved with forward substitution for $\{D\}^T = [1 \quad -0.03333 \quad -0.1009]$

$$\begin{bmatrix} 3 & -0.1 & -0.2 \\ 0 & 7.00333 & -0.293333 \\ 0 & 0 & 10.0120 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 1 \\ -0.03333 \\ -0.1009 \end{Bmatrix}$$

solved by back substitution
 $\{X\}^T = [0.33249 \quad -0.00518 \quad -0.01008]$

 U

Matrix Inverse from LU Decomposition

The first column of the inverse matrix

$$[A]^{-1} = \begin{bmatrix} 0.33249 & 0 & 0 \\ -0.00518 & 0 & 0 \\ -0.01008 & 0 & 0 \end{bmatrix}$$

To determine the second column

$$\begin{bmatrix} 1 & 0 & 0 \\ 0.0333333 & 1 & 0 \\ 0.100000 & -0.0271300 & 1 \end{bmatrix} \begin{Bmatrix} d_1 \\ d_2 \\ d_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 1 \\ 0 \end{Bmatrix} \quad \text{The second column of the inverse matrix}$$

$\{X\}^T = [0.004944 \quad 0.142903 \quad 0.00271]$

$$[A]^{-1} = \begin{bmatrix} 0.33249 & 0.004944 & 0 \\ -0.00518 & 0.142903 & 0 \\ -0.01008 & 0.00271 & 0 \end{bmatrix}$$

With $\{B\}^T = [0 \quad 0 \quad 1]$

$$\{X\}^T = [0.006798 \quad 0.004183 \quad 0.09988]$$

Assignment:

Given $x_1 = -2$, $x_2 = -1$, $x_3 = 1$ and $x_4 = 2$

1. A 4x4 $|A|$ and a 4x1 $|b|$ matrix like below (or you may make)
2. Make L and U matrix from A using LU decomposition method
3. Show that the values x like above.

$$\begin{vmatrix} 12 & -3 & 6 & 4 \\ 2 & 11 & 3 & 5 \\ 4 & 1 & 10 & 2 \\ -5 & 3 & 4 & -10 \end{vmatrix} \cdot \begin{vmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{vmatrix} = \begin{vmatrix} -7 \\ -2 \\ 5 \\ -9 \end{vmatrix}$$